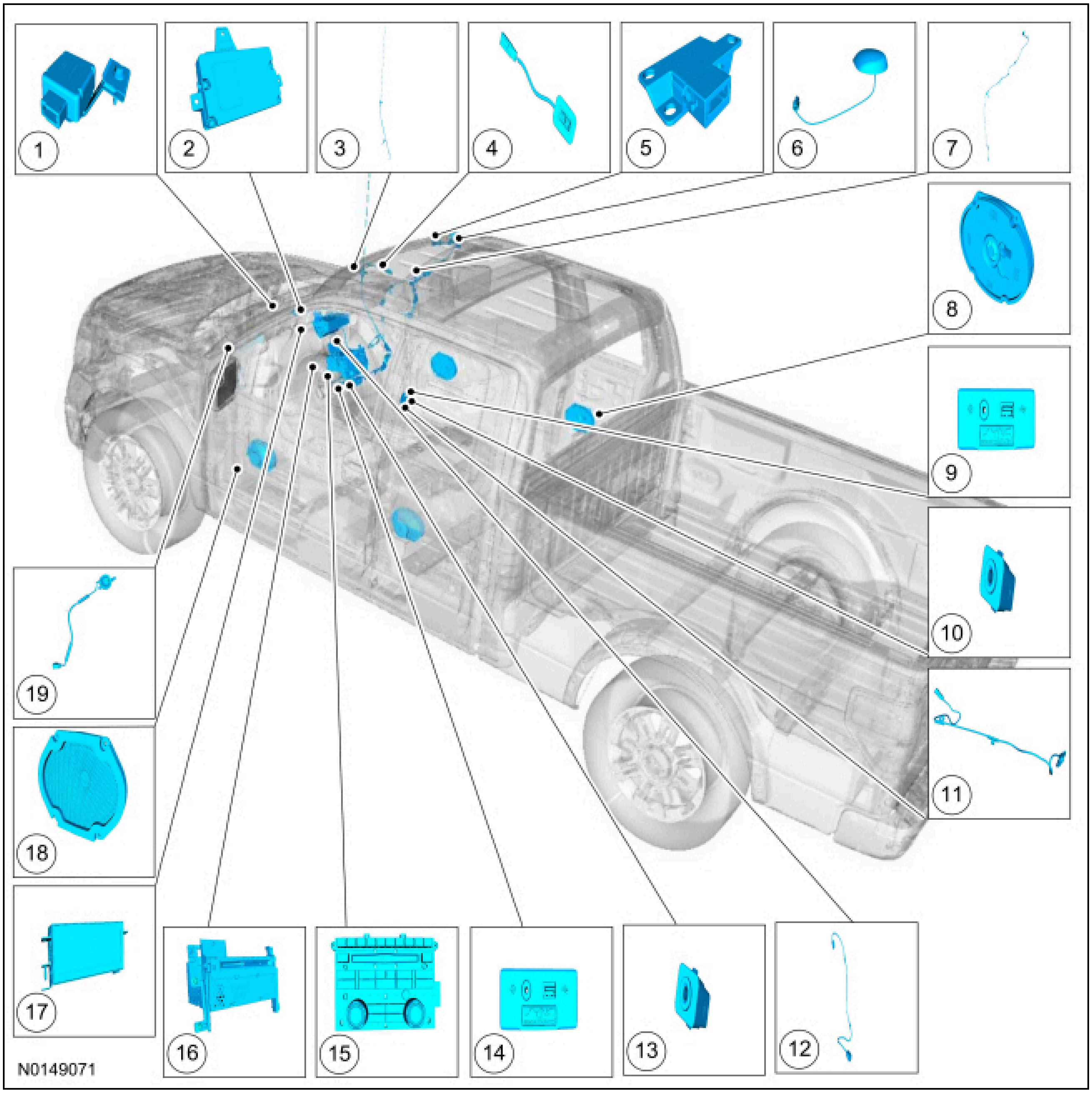
Information and Entertainment System

Ford Ecat Electronic Parts Catalog

Component Location



Item	Description	Comments
1	GPSM	Located in the top center area of the instrument panel (if equipped)
2	APIM	Located in the top center area of the instrument panel (if equipped)
3	AM/FM antenna and AM/FM antenna coaxial	-

Item	Description	Comments
	cable	
4	SYNC® microphone	Located in the headliner (if equipped)
5	Compass module	Located in the headliner (if equipped)
6	Satellite radio antenna	If equipped
7	Satellite radio antenna coaxial cable	If equipped
8	Rear door speaker	-
9	Audio input jack and USB port	Located in the console (with front captain chairs)
10	Audio input jack	Located in the console (with front captain chairs)
11	USB cable	Located in the console (with front captain chairs)
12	USB cable	Located in the instrument panel (with split bench)
13	Audio input jack	Located in the instrument panel (with split bench)
14	Audio input jack and USB port	Located in the instrument panel (with split bench)
15	FCIM	-
16	ACM	-
17	Non-touchscreen FDIM	-
18	Front door speaker	-
19	A-pillar speaker	LF shown, RF similar

Overview

The audio system consists of the following:

Main Components

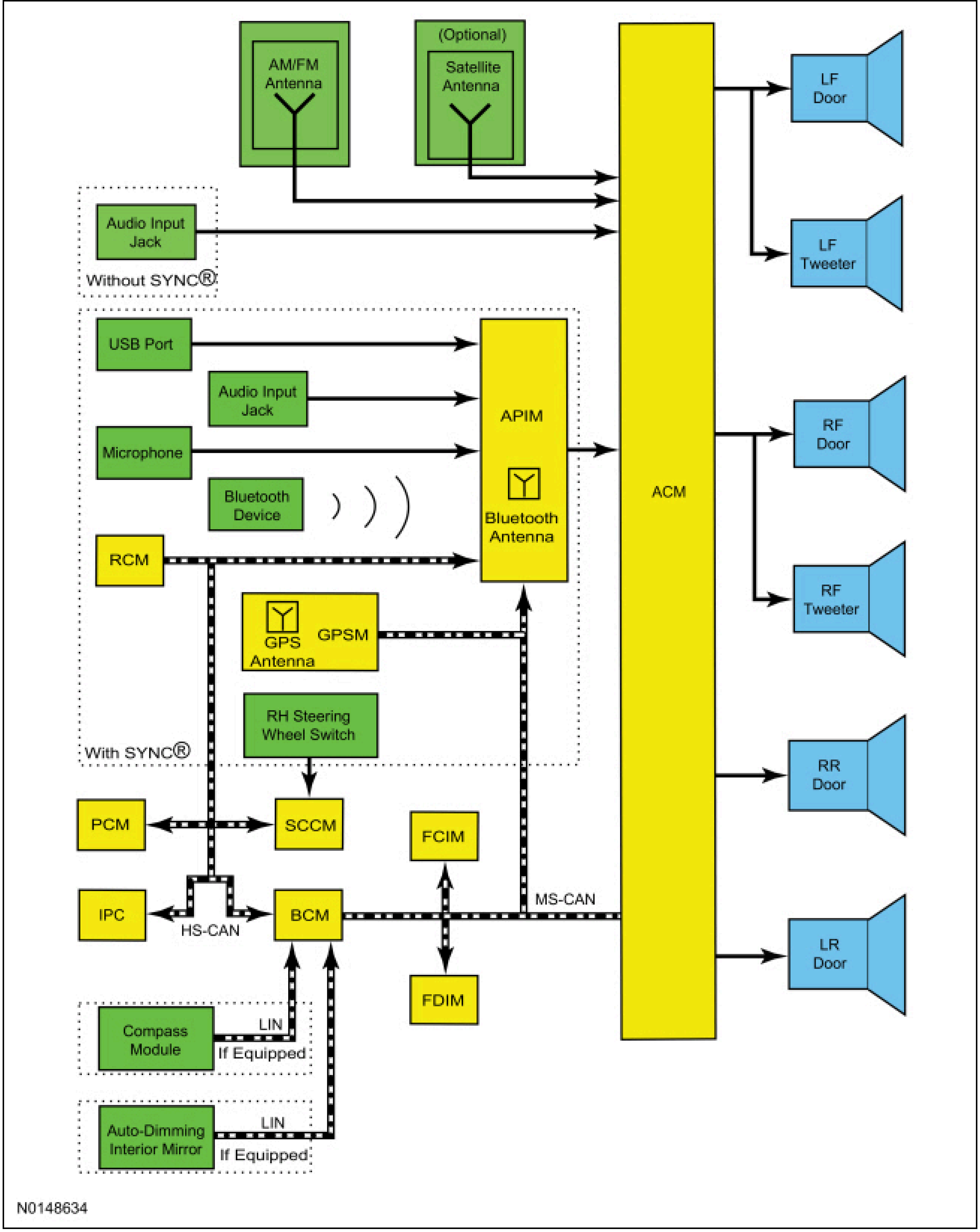
- 6 speakers
- AM/FM single CD ACM
- APIM (with SYNC®)
- Audio input jack
- Compass module (if equipped)
- FCIM
- GPSM (with SYNC®)
- Auto-dimming interior rear view mirror (if equipped)
- Non-touchscreen FDIM
- Redundant steering wheel controls (with SYNC®)
- SYNC® microphone (with SYNC®)
- USB port (with SYNC®)

Main Features

- Optional satellite radio
- Optional SYNC®

System Operation

System Diagram



Network Message Chart

NOTE: *A missing network message is indicated by a U-code DTC . When a missing message DTC (U-code) sets, it is important to look for symptoms present in the audio system and throughout the vehicle, and to review the complete message list to determine which other modules rely on the same message. Refer to CAN Module Communication Message Chart in Section 418-00. When the other modules have been identified, run the self-test for those modules. If the same message is missing from those modules, the identical, or similar lost communication DTCs may be set in those modules. Confirmation of missing messages common to multiple modules can indicate that the originating module is the source of the concern, or that the communication network may be at fault.*

It is very important to understand the following:

- *Where the input originates*
- *All of the information necessary for a feature to operate*
- *Which module(s) receive(s) the input or command message*
- *Which module controls the output of the feature*
- *If the module that receives the input controls the output of the feature, or whether it outputs a message over the CAN to another module.*

Module Network Input Messages - ACM

Broadcast Message	Originating Module	Message Purpose
Accessory delay status	BCM	Used to determine the audio system operating mode.
Ignition switch position	BCM	Used to indicate the ignition state (OFF, ACCESSORY, RUN, and START) required for ACM operating modes and fault reporting.
Steering wheel switch media	SCCM	Used to set the audio system listening mode (AM/FM, CD, etc.) from the steering wheel switch.
Steering wheel switch seek left	SCCM	Used to access the next strong station to the left, to restart an audio track, to access the previous satellite radio station preset (if equipped), or the previous audio track from the steering wheel switch.
Steering wheel switch seek right	SCCM	Used to access the next strong station to the right, to access the next satellite radio station preset (if equipped), or to access the next audio track from the steering wheel switch.
Steering wheel switch volume down	SCCM	Used to decrease the audio system volume from the steering wheel switch.
Steering wheel switch volume up	SCCM	Used to increase the audio system volume from the steering wheel switch.
Vehicle speed	PCM	Used for the speed compensated volume function.

Module Network Input Messages - APIM

Broadcast Message	Originating Module	Message Purpose
Accessory delay status	BCM	Used to determine the audio system operating mode.
Airbag deployment eCall notification	RCM	Used to notify of a 911 assist call being initiated due to airbag deployment.
GPS location	GPSM	Used for vehicle positioning, heading, and direction for turn by turn navigation and 911 assist.
Steering wheel switch OK	SCCM	Used to confirm various audio system selections.
Steering wheel switch phone	SCCM	Used to answer or end a phone call from the steering wheel switch.

Broadcast Message	Originating Module	Message Purpose
Steering wheel switch voice	SCCM	Used to access or exit the voice command feature from the steering wheel switch.

Module Network Input Messages - FDIM

Broadcast Message	Originating Module	Message Purpose
Compass display data	BCM	Used to display the compass direction on the FDIM .
Compass zone and calibration status	BCM	Used to display compass zone and calibration status on the FDIM .

Module Network Input Messages - GPSM

Broadcast Message	Originating Module	Message Purpose
Ignition switch position	BCM	Used to indicate the ignition state (OFF, ACCESSORY, RUN, and START) required for GPSM operating modes and fault reporting.
Selector lever (PRNDL) status	BCM	Used to verify the gear lever position for vehicle direction data.

Accessory Delay Feature

The accessory delay feature allows the audio system to be operated for a preset period of time after the ignition is turned off and a front door has not been opened.

AM/FM Radio

The AM/FM fender-mounted antenna receives radio waves and sends them through the AM/FM antenna coaxial cable to the ACM . The radio waves are converted into fluctuating AC voltage and sent as left and right analog audio signals to the speakers.

Audible Prompts

The APIM receives stereo inputs from connected mobile phones and mono inputs from the SYNC® microphone.

Stereo and mono inputs, voice prompts, the TTS feature, ringtones, and any audio received through a connected mobile phone are transmitted to the ACM as stereo or mono sound.

The TTS feature speaks information so that it does not have to be read from the display.

Battery Load Shed

The BCM uses the battery current sensor to monitor the battery state of charge. The battery current sensor is attached to the battery ground cable. With the vehicle OFF and the ignition in ACC or ON, a load shed message is sent over the CAN when the BCM determines that the battery state of charge is below 40%, 45 minutes have elapsed, or 10% of the charge has been drained. This message turns off the audio system to save the remaining battery charge. Under this condition, SYS OFF TO SAVE BATT is displayed on the centerstack infotainment display to notify the driver that battery protection actions are active. To clear the load shed state, start the vehicle.

Bluetooth Mode

Bluetooth is a secure, short-range radio frequency that allows devices to communicate wirelessly through radio waves. The operating range of a Bluetooth signal is a maximum of 9.75 m (32 feet).

The Bluetooth interface can accommodate both Bluetooth-enabled mobile phones and Bluetooth-enabled media devices. The SYNC® system allows interaction with several types of customer Bluetooth devices, including mobile phones and media devices. The APIM contains an on-board Bluetooth chipset, which enables certain wireless devices to interact with the system. Any Bluetooth device used with the SYNC® system must first be paired with the system before it is operational.

Only one Bluetooth phone and one Bluetooth media device can be connected to the system at any one time. If an additional device of either type is paired with the system and made active, the APIM disconnects any active connection and establishes a connection with the new device.

When a new Bluetooth device is added, the APIM and the Bluetooth device must be paired together. Most Bluetooth devices can pair with the SYNC® system, although functionality may vary. To determine if a Bluetooth device is supported, verify the customer device is on the compatibility list for the current APIM software level.

Pairing a Bluetooth device is accomplished through the "Add Device" selection of the phone menu. When pairing a device, the SYNC® system generates a unique PIN that must be entered on the Bluetooth device in order for the pairing process to be successful. There are also some device-specific actions that must take place. For information on the pairing process, refer to the Owner's Literature.

It is important to understand that not all mobile phones have the same level of features when interacting with the SYNC® system. For a list of compatible phones, refer to SyncMyRide website.

MyKey® Audio Operation

When a MyKey® is in use, the following restrictions are activated:

- The audio system is muted until the safety belts are buckled when the Belt-Minder® feature is activated. The MyKey® "BUCKLE UP TO UNMUTE AUDIO" is displayed in the information display.
  - This is a standard setting and cannot be changed.
- Satellite radio adult content stations are blocked by being muted.
- If "Always On" is selected in the MyKey® menu, 911 assist and the Do Not Disturb features are enabled and cannot be disabled.
- The audio system volume is limited to 45%. In an attempt to exceed the limited volume, the MyKey® VOLUME LIMITED message is displayed in the centerstack infotainment display.
  - This is an optional setting.

Refer to the Owner's Literature for more information on MyKey®.

Satellite Radio (If Equipped)

Satellite radio is a subscription-based service.

The satellite radio roof-mounted antenna receives signals containing data (station call letters, artist names, song titles, etc.) and digital satellite radio audio. These signals are sent through the satellite radio antenna coaxial cable to the satellite radio receiver built into the ACM . The ACM separates the data from the audio. The data is sent to be displayed on the centerstack infotainment display, and the audio is converted into fluctuating AC voltage and sent as left and right analog audio signals to the speakers.

Speed Compensated Volume

The ACM adjusts the audio system volume based on the VSS signal to compensate for speed and wind noise.

When a MyKey® is in use and the MyKey® radio volume limiter is enabled, the speed compensated volume is disabled.

Steering Wheel Switch Functions

The RH 8-way steering wheel switches operate the volume +/-, seek +/-, and the audio system SYNC® functions (media, voice, phone, and OK).

SYNC® AppLink™

SYNC® AppLink™ allows the customer to control smartphone applications by using voice commands and in-vehicle controls. Data is transferred from the customer's device to the APIM through Bluetooth, or through the USB port with Apple® devices.

SYNC® Services

SYNC® Services is a subscription-based service. For SYNC® Services subscription information,, refer to SyncMyRide.

The customer's mobile phone is used to access and download the SYNC® services. SYNC® services information received by the customer's mobile device is sent to the APIM .

The GPS antenna, integral to the GPSM , is used to detect the location and direction of the vehicle.

SYNC® services includes the following features:

- Business search
- Personalized text alerts
- Real-time traffic updates
- Turn-by-turn directions

SYNC® services allows the customer to receive updates for:

- Entertainment
- News
- Sports
- Stocks
- Travel
- Weather

SYNC® System

The SYNC® system is a hands-free communication and entertainment system that supports many features. All features are not available with every phone.

The SYNC® system provides the ability to:

- Connect media devices (such as an iPod® or USB device) in order to play audio files.
- Control the entertainment system using voice controls.
- Initiate an emergency call (911 Assist™) when the airbags deploy.
- Play media files via a paired Bluetooth-enabled audio device.
- Produce vehicle health reports.
- Send and receive phone calls via a paired Bluetooth-enabled phone.
- Send and receive text messages via a paired Bluetooth-enabled phone.

USB Audio Mode

The USB port can be used for connecting a media device (such as an iPod®) with the device's available cable, or for directly plugging in a portable mass storage device, such as a thumb drive. When playing media files stored on a mass storage device, the SYNC® system only plays files that do not have DRM protection. The USB port can also be used for uploading vehicle application upgrades. The USB port is powered by the APIM , so no external power source is needed to power a device plugged into the USB port if the device supports this feature.

In addition to audio information, metadata (information such as artist, album title, song title, and genre) may also be sent to the APIM from a device plugged into the USB port. The APIM uses the metadata to create indexes that can be used to sort for particular music, based on customer preference. Not all USB devices can send metadata to the APIM . When a new media device is connected to the SYNC® system, the APIM automatically indexes the information. This may take several minutes (depending on the amount of data on the device), and is considered normal operation. When a device that was previously connected to the SYNC® system is reconnected, the APIM updates the index (rather than creating a new one), which reduces the amount of time needed to index the device.

Voice Recognition

Voice recognition is used for many vehicle functions, including the audio system. The microphone relays the microphone input to the APIM through dedicated wiring. The TTS and voice prompt features speak certain text information and interaction requests to minimize driver distraction by having to look at the audio system display while driving. The ringtone alerts the driver to an incoming call. The microphone is also used to detect outgoing audio during a phone call.

Audible prompts can range from a simple tone to more elaborate spoken text, based on the customer setting. When interaction mode is set to standard, detailed guidance is provided. When interaction mode is set to advanced, most prompts are tones only and minimal audible guidance is provided. Refer to the Owner's Literature for further information on voice interaction.

The audio signals for the TTS and voice prompt features, ringtones, and audio from the outside device during a phone call, are sent from the APIM to the ACM .

Component Description

ACM

The ACM can be operated with the ignition in RUN, ACC, or OFF.

The ACM receives radio waves containing audio from the antenna coaxial cable(s) and converts them into fluctuating AC voltage. The ACM receives audio signals from the APIM in the form of fluctuating AC voltage. The ACM processes these inputs and sends left and right analog audio signals to the speakers as fluctuating AC voltage.

If equipped with satellite radio, the ACM will also receive radio waves containing data through the antenna coaxial cable(s). The data is sent to be displayed on the centerstack infotainment display.

The ACM requires PMI when it is replaced.

Antenna

The AM/FM fender-mounted antenna receives radio waves containing audio and sends them through the AM/FM antenna coaxial cable to the ACM . The antenna mast is used to amplify the AM/FM radio waves to improve reception.

If equipped with satellite radio, the satellite radio roof-mounted antenna receives signals containing data and digital audio. These signals are sent through the satellite radio antenna coaxial cable to the satellite radio receiver built into the ACM .

APIM

Audio signals received by the APIM are sent to the ACM as fluctuating AC voltage.

The APIM consists of 2 internal modules: the CIP and the VIP . The modules are not replaceable individually, but can be flashed independently, if required.

The CIP interfaces with all of the inputs to the APIM . It contains an analog-to-digital-to analog converter, as well as the Bluetooth chipset. Any Application upgrades that are available to the consumer are loaded directly to the CIP through the USB port.

The VIP provides an interface between the CIP and the vehicle. Its main functions are controlling the APIM power management and translating inbound and outbound CAN signals. In addition, the VIP queries the modules on the network to retrieve any DTCs when a vehicle health report is requested.

Audio Input Jack

The audio input jack allows for a portable MP3 audio device (such as an iPod®) to be connected to the vehicle audio system utilizing a 1/8-inch (3 mm) audio jack.

Auto-Dimming Interior Rear View Mirror (If Equipped)

If the vehicle is equipped with an auto-dimming interior rear view mirror, a compass module will not be used.

The auto-dimming interior rear view mirror communicates the vehicle direction to the BCM through a LIN circuit. The data is then sent over the MS-CAN to the FDIM , where it is displayed. The BCM is only a gateway for the compass information and does not control the compass display

Compass Module (If Equipped)

The vehicle is eqiupped with a compass module if an auto-dimming interior rear view mirror is not used.

The compass module communicates the vehicle direction to the BCM through a LIN circuit. The data is then sent over the MS-CAN to the FDIM , where it is displayed. The BCM is only a gateway for the compass information and does not control the compass display

FCIM

The FCIM can be operated with the ignition in RUN, ACC, or OFF.

The FCIM audio system buttons provide one of the methods in which the customer interacts with the infotainment system.

The FCIM is separate from the ACM .

FDIM

The FDIM receives CAN messages for all of its displays, which include:

- Audio information
- Climate control fan speed and temperature setting
- Outside air temperature
- Compass heading
    - The magnetic fields generated by the earth's north and south poles are divided into zones that differ from each other with respect to how the magnetic fields appear. The magnetic north can change by up to 4 degrees between zones and become noticeable across multiple zones. The compass zone must be set to the correct zone for the compass to read accurately. Factors within the vehicle, and outside factors such as bridges, steel structures, tall buildings, and power lines may affect compass accuracy. Calibrating the compass allows the auto-dimming interior rear view mirror (if equipped) or compass module (if equipped) to compensate for vehicle factors that influence the accuracy of the compass heading.

Global Positioning System Module (GPSM)

The Global Positioning System Module (GPSM) provides vehicle location for real-time traffic reports and re-routing, and for identifying vehicle location in the event of a collision.

Vehicle location information is broadcast from the Global Positioning System Module (GPSM) to the APIM .

Steering Wheel Switches

The RH steering wheel switches contain series of resistors. Each steering wheel switch function corresponds with a specific resistance value within the switch.

The SCCM sends a 5-volt reference voltage to the RH steering wheel switch on the input circuits. When a switch is pressed, the voltage is routed through the corresponding resistor(s), and then to ground through the SCCM . The SCCM monitors the resultant voltage drop to determine which switch was pressed. The voltage drop varies depending upon the resistance of the specific switch pressed.

The SCCM sends a network message to the APIM for the SYNC® switches and to the ACM for the redundant audio controls.

SYNC® Microphone

The SYNC® microphone receives SYNC® system voice commands and outgoing audio during a phone call. These signals are sent to the APIM .

USB Port

The USB port is connected to the APIM through the USB cable. The USB port can be used to play audio files, connect a media device, or upload software from mass storage.